HR AI Explainer

Predicting and Understanding Turnover through Explainable & Ethical AI

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The Challenge of Modern Turnover

High Cost of Attrition

Employee turnover represents significant financial and operational costs for organizations, requiring proactive identification of at-risk employees.

Regulatory Compliance

AI Act mandates algorithmic transparency and fairness audits to ensure ethical deployment in HR decision-making processes.

Resource Efficiency

Need for computationally efficient models with minimal carbon footprint that deliver actionable insights without excessive infrastructure costs.

Research Question

How can we design a frugal analytical model capable of identifying, from HR data, the factors contributing to employee turnover risk, while ensuring an ethical and explainable approach?

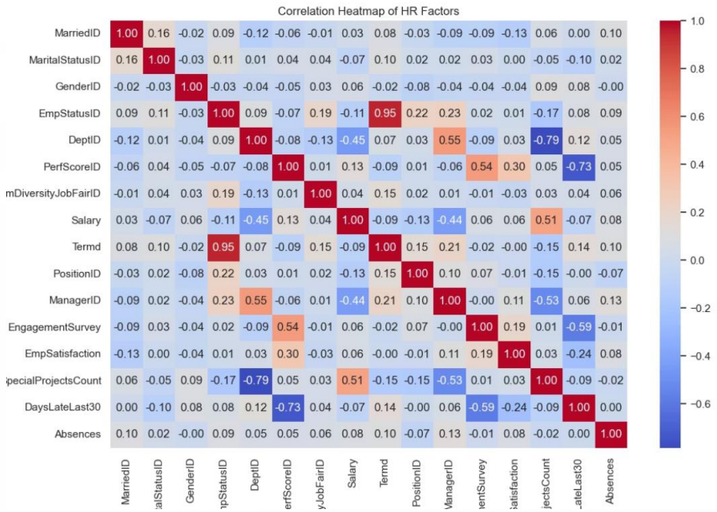
Understanding the Human Data

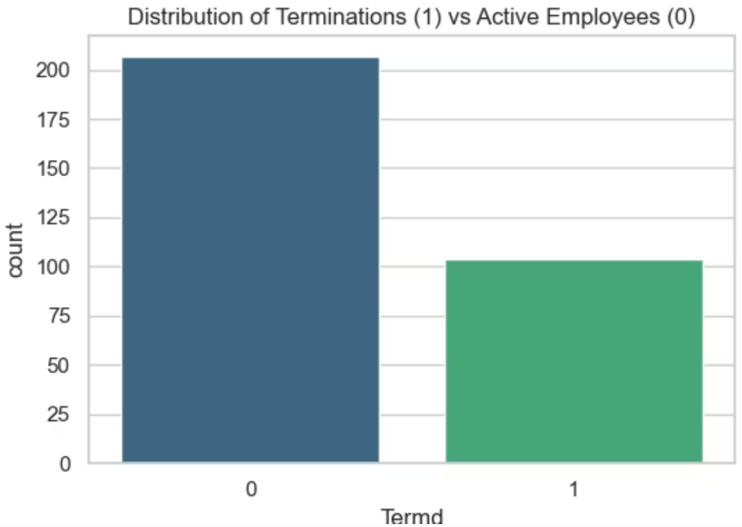
Analysis of HR_anonymized.csv: Identification of critical behavioral patterns among employees.

Key correlation: Highlighting significant links between satisfaction, salary, and tenure.

Predictive analytics: Revealing hidden relationships through the correlation matrix.

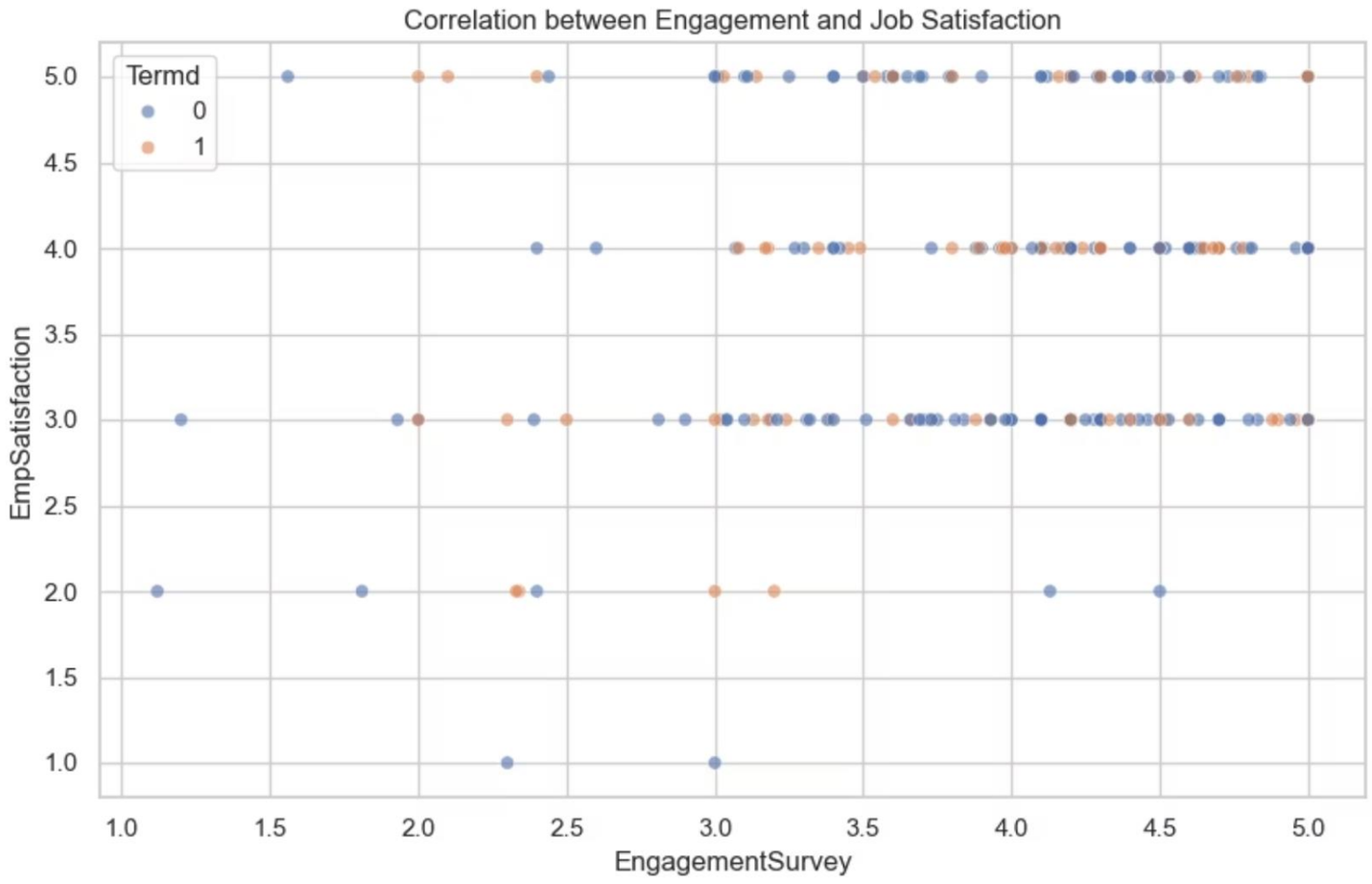
Anonymisation: It removes columns containing personally identifiable information such as employee names (Employee_Name), identifiers (EmpID), manager names, and postal codes.





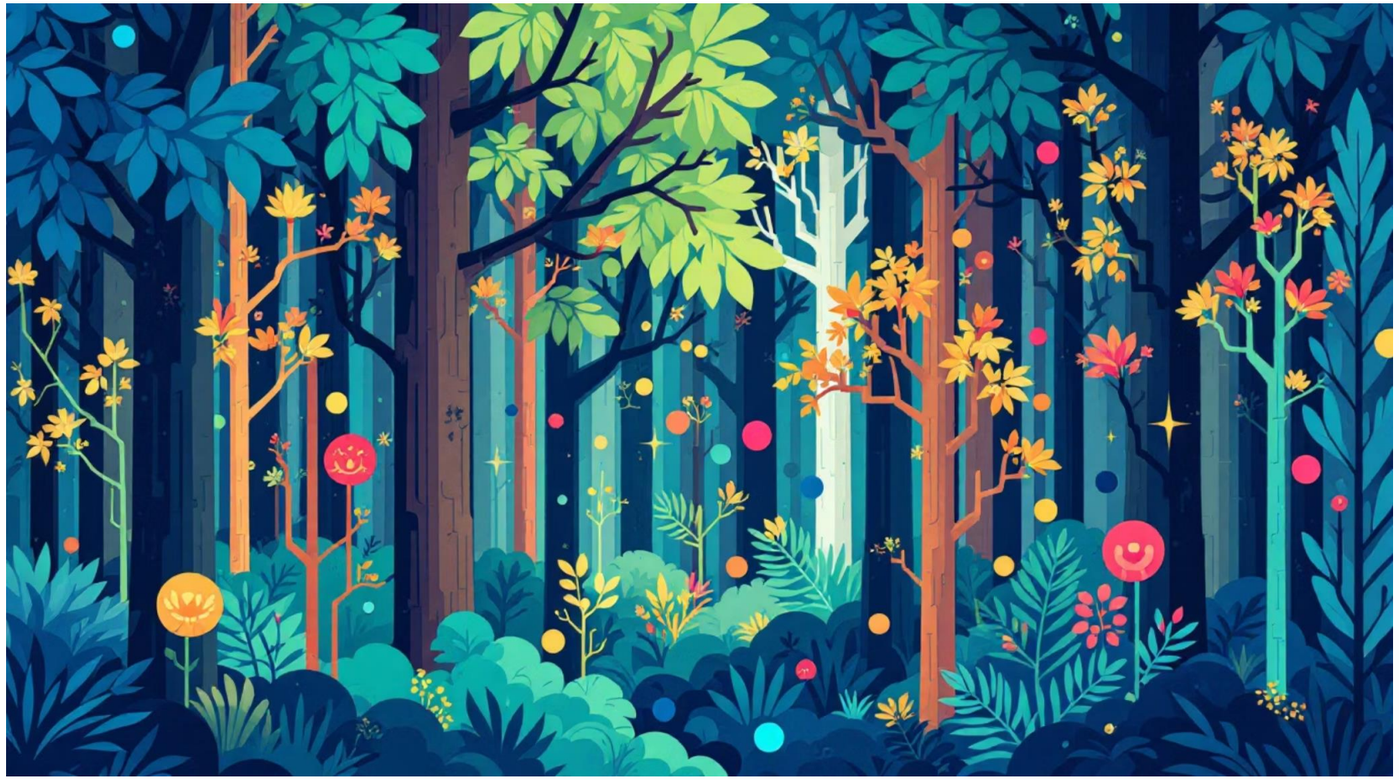
The Threshold of Retention

Visual Proof of Impact



- **Algorithmic Bias Monitoring**: Automatically detects potential discrimination to ensure fair treatment across all employee groups.
- **Regulatory Compliance (80% Rule)**: Validates the model against legal standards (Four-Fifths Rule) to guarantee neutral and unbiased predictions.

A Frugal & Robust Model: Random Forest

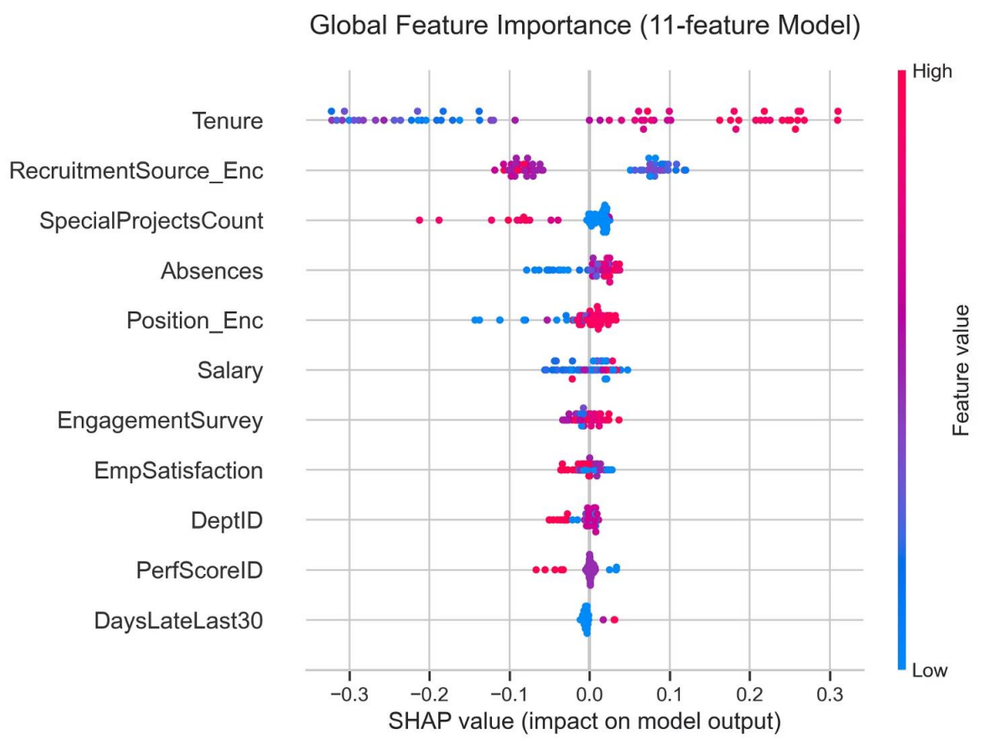


Why Random Forest?

- High prediction accuracy with minimal overfitting
- Low computational cost and fast training time
- Natural handling of non-linear relationships
- Intrinsic feature importance ranking

Carbon footprint tracking via CodeCarbon revealed minimal emissions (referencing emissions_ml.csv), making it ideal for environmentally conscious AI deployment.

Data Processing & Model Optimization



Data Processing

Calculation of seniority, encoding of categories, and class balancing via SMOTE method (to address the lack of departure data).

Optimization

Search for the best hyperparameters for a Random Forest model through cross-validation.

Performance

The model achieves a final overall accuracy of approximately 73%.

Explainability (SHAP)

It generates SHAP values to understand which factors (salary, seniority, etc.) most influence predictions.

Fair AI

0.995

Disparate Impact Score

Excellent fairness metric indicating no significant bias across protected attributes

100%

80% Rule Compliance

Fully compliant with regulatory requirements for algorithmic fairness

Comprehensive fairness audit conducted using AIF360 framework (referencing 3_Audit_Etique.ipynb) confirmed the model's ethical deployment. The analysis examined gender, age, and other protected attributes, ensuring unbiased predictions across all demographic groups.

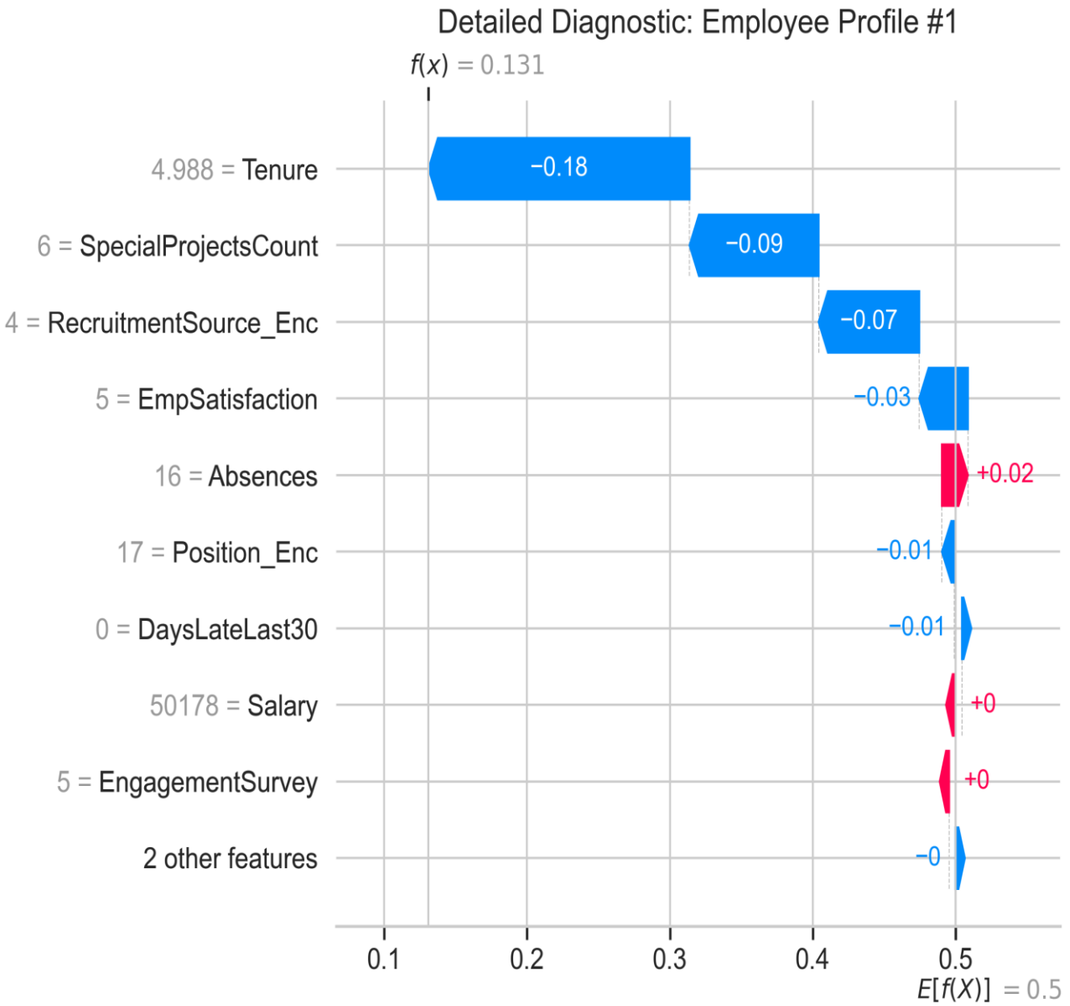
Compliance Status: The model is COMPLIANT and unbiased regarding gender and protected attributes (80% rule).

From Statistics to Individual Insights

Waterfall Effect in Action

The waterfall visualization demonstrates how individual features combine to create a complete risk profile for a single employee. Each bar represents how a specific factor adds to or subtracts from the baseline risk score.

HR managers can trace the complete logic path from raw data to final prediction, understanding exactly why an employee is flagged as at-risk. This transparency builds trust and enables informed intervention decisions.



High-Performance & Modular Backend

```
rh_ai_explainer/
 ├── app/
 |   ├── main.py            # Fast API application
 |   ├── predictor.py       # ML prediction logic
 |   ├── ollama_client.py   # LLM integration
 |   ├── prompts.py         # Prompt templates
 |   ├── schemas.py         # Pydantic schemas
 |   ├── settings.py        # Configuration
 |   └── templates/
 |       └── index.html     # Frontend UI
 ├── HR_anonymized.csv      # HR Dataset
 ├── rf_model.joblib        # Trained Random Forest model
 ├── shap_explainer.joblib  # SHAP explainer
 ├── le_source.joblib       # Label encoder for recruitment sources
 ├── le_pos.joblib          # Label encoder for positions
 └── requirements.txt       # Python dependencies
```

FastAPI Framework

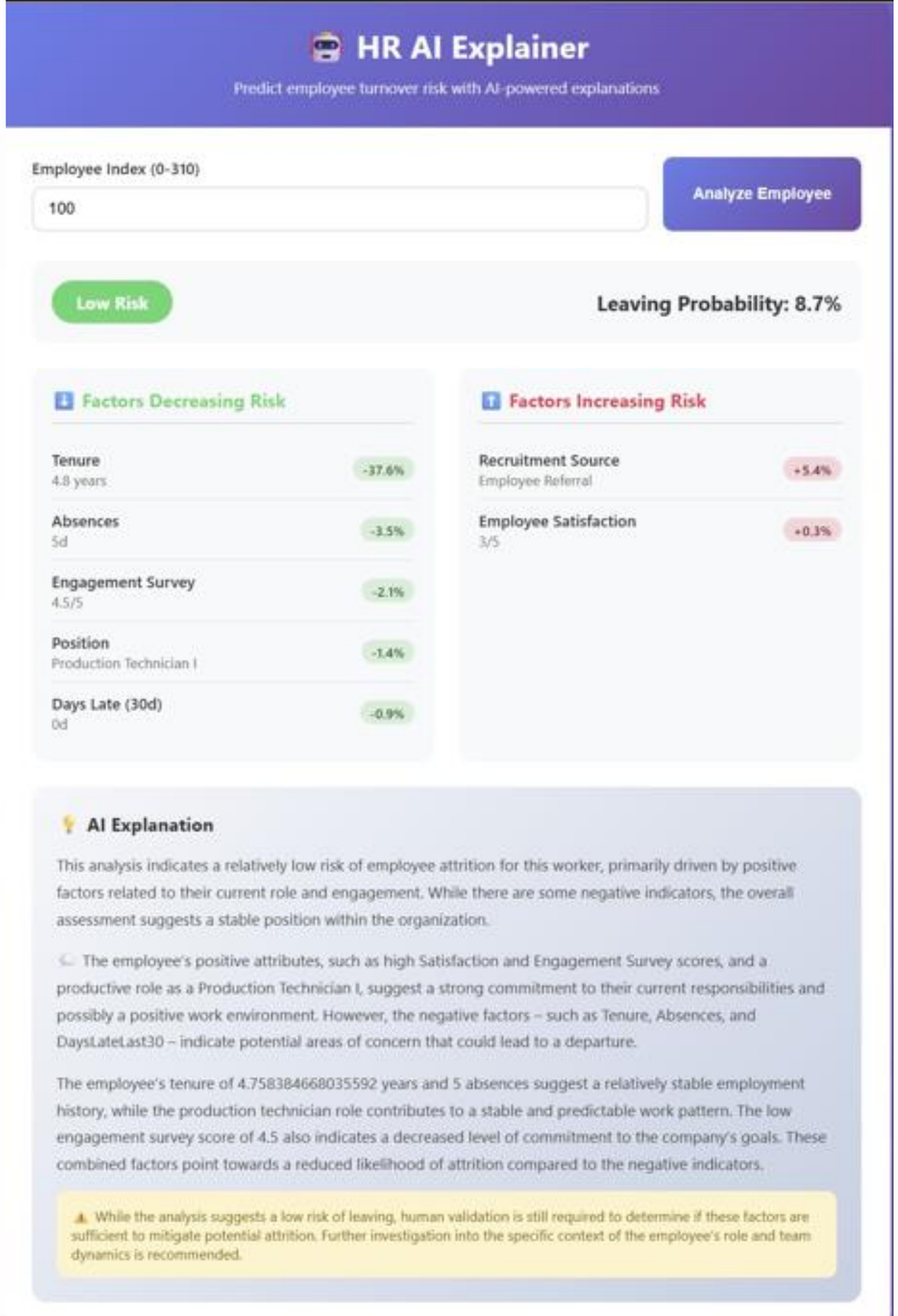
High-speed asynchronous API enables real-time predictions with sub-second response times, supporting concurrent requests from multiple HR managers.

Pre-Trained Models

Joblib serialization enables instant model loading without retraining, delivering immediate predictions upon system startup.

Modular Architecture

Separate components for predictor.py, ollama_client.py, and shap_explainer ensure maintainability and scalability as requirements evolve.



Empowering HR Managers

User-Centric Frontend Design

Intuitive interface combines risk exploration with natural language explanations powered by Gemma 3 via Ollama. Automatic translation of SHAP values into plain English enables non-technical users to understand complex predictions.

Clean UI

Minimalist design prioritizes actionable insights over clutter, focusing on what matters most to HR decision-makers.

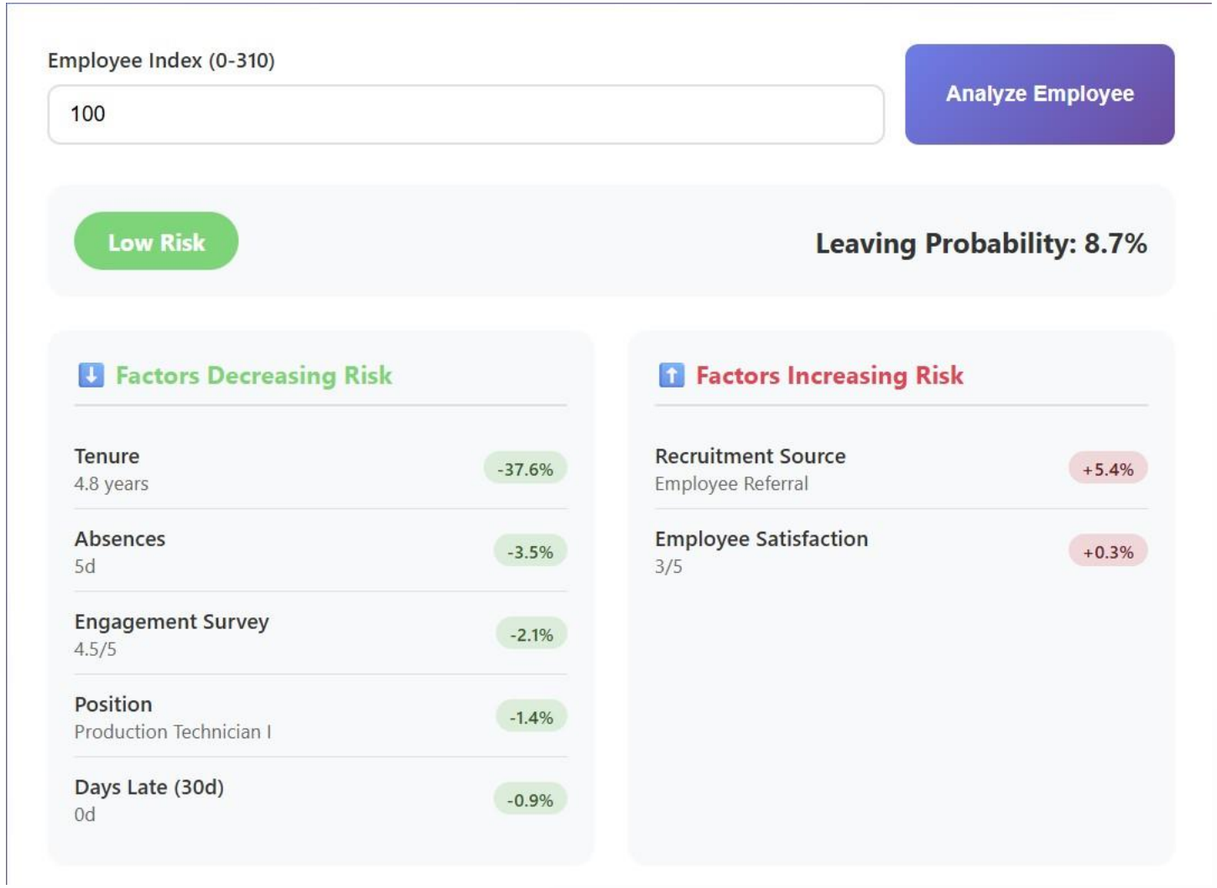
Natural Language

Gemma 3 transforms technical SHAP parameters into clear, conversational explanations anyone can understand.

Visual Analytics

Interactive charts and graphs reveal patterns instantly, supporting data-driven conversations between HR and leadership.

Frontend page



ID-Based Risk Retrieval: Instantly access any employee's profile to get a real-time turnover probability score.

Point-by-Point SHAP Analysis: Breaks down the prediction into specific drivers, showing exactly which factors increase or decrease the risk

From Black-Box to Transparent AI: Converts complex algorithmic results into clear, justifiable reasons for HR decision-makers.

Targeted Retention Insights: Identifies the primary causes of potential departures (e.g., salary, tenure) to enable personalized action plans.

Frontend page

💡 **AI Explanation**

This analysis indicates a relatively low risk of employee attrition for this worker, primarily driven by positive factors related to their current role and engagement. While there are some negative indicators, the overall assessment suggests a stable position within the organization.

💭 The employee's positive attributes, such as high Satisfaction and Engagement Survey scores, and a productive role as a Production Technician I, suggest a strong commitment to their current responsibilities and possibly a positive work environment. However, the negative factors – such as Tenure, Absences, and DaysLateLast30 – indicate potential areas of concern that could lead to a departure.

The employee's tenure of 4.758384668035592 years and 5 absences suggest a relatively stable employment history, while the production technician role contributes to a stable and predictable work pattern. The low engagement survey score of 4.5 also indicates a decreased level of commitment to the company's goals. These combined factors point towards a reduced likelihood of attrition compared to the negative indicators.

⚠️ While the analysis suggests a low risk of leaving, human validation is still required to determine if these factors are sufficient to mitigate potential attrition. Further investigation into the specific context of the employee's role and team dynamics is recommended.

AI-Powered Strategic Advisor: Converts complex data and SHAP values into clear, natural language recommendations for managers.

Personalized Action Plans: Generates tailored advice (like salary reviews or career coaching) based on each employee's unique profile.

Bridging Data & HR Intuition: Harmonizes technical machine learning outputs with human-centric HR management strategies.

Trustworthy AI: Security & Sustainability

Cybersecurity
Data anonymization and strict protection against Prompt Injection

Frugality
100% local deployment using a lightweight LLM (Ollama). No cloud dependency or data leakage.

Reliability
Low temperature setting for the LLM to ensure factual, non-creative, and consistent explanations.

Explainability
Every prediction is justified by SHAP parameters to avoid "Black Box" decisions.

Conclusion & Future Impact

Key Value Pillars

Balanced Innovation
Achieving the perfect equilibrium between **high-performance forecasting**, **environmental sobriety**, and **total regulatory compliance**.

The End of the "Black Box“
Transforming complex data into a transparent, trust-based dialogue between the organization and its employees.

Strategic Outcomes

Optimized Retention
Shift from reactive to proactive HR management with data-driven, personalized career strategies.

Ethical Leadership
Setting a new standard for HR Tech by integrating **Fairness-by-Design** and **Green IT** (Carbon-tracked AI) into the corporate DNA.